

AD2V33**CYBER SECURITY**

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COURSE OBJECTIVES:

- To learn cybercrime and cyber law.
- To understand the cyber-attacks and tools for mitigating them.
- To understand network security attacks.
- To learn how to detect and prevent a cyber-attack.
- To learn about ethical hacking.

UNIT I INTRODUCTION

9

Cyber Security – History of Internet – Impact of Internet – CIA Triad – Cyber Physical System; Reason for Cyber Crime – Need for Cyber Security – History of Cyber Crime; Cybercriminals – Classification of Cybercrimes – A Global Perspective on Cyber Crimes; Cyber Laws – The Indian IT Act – Cybercrime and Punishment.

UNIT II ATTACKS AND COUNTERMEASURES

9

Malicious Attack Threats and Vulnerabilities: Scope of Cyber-Attacks – Security Breach – Types of Malicious Attacks – Malicious Software – Common Attack Vectors – Social engineering Attack – Web Application Attack – Attack Tools – Countermeasures.

UNIT III ATTACKS ON NETWORK

9

Network concepts – Threats to network communication – Wireless network security – Denial of Service – Distributed Denial of Service – Cryptography in Network Security.

UNIT IV INTRUSION DETECTION AND PREVENTION

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Host -Based Intrusion Detection – Network -Based Intrusion Detection – Distributed or Hybrid Intrusion Detection – Intrusion Detection Exchange Format – Honeypots – Example System Snort – Types of Firewalls and Intrusion Prevention Systems.

UNIT V ETHICAL HACKING

9

Information gathering - Open-source intelligence gathering – Windows: Netcraft- Linux: Whois Lookup - Password attack - Online Password attacks – Offline password attacks – Wireless attacks.

TOTAL: 45 PERIODS**COURSE OUTCOMES:**

At the end of this course, the students will be able to:

- CO1:** Explain the basics of cyber security, cyber-crime and cyber law.
- CO2:** Classify various types of attacks and learn the tools to launch the attacks.
- CO3:** Apply various tools to perform information gathering.
- CO4:** Apply intrusion techniques to detect and prevent intrusion.
- CO5:** Apply concepts of ethical hacking.

TEXT BOOKS:

1. Charles Pfleeger, Shari Lawrence Pfleeger, Lizzie Coles-Kemp, "Security in Computing", Sixth Edition, Addison-Wesley Professional, 2023 (Unit 3)
2. William Stallings, Lawrie Brown, "Computer Security Principles and Practice", Third Edition, Pearson Education, 2015 (Units 4)

REFERENCES :

1. Anand Shinde, "Introduction to Cyber Security Guide to the World of Cyber Security", Notion Press, 2021 (Unit 1)
2. Nina Godbole, Sunit Belapure, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley Publishers, 2011 (Unit 1)
3. David Kim, Michael G. Solomon, "Fundamentals of Information Systems Security", Jones & Bartlett Learning Publishers, 2013 (Unit 2)
4. Georgia Weidman, "Penetration Testing: A Hands-On Introduction to Hacking", No Starch Press, 2014 (unit 5)